CARLOS HOMERO JUCA VITERI

Software Engineer | Full Stack · Cloud · AI

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PROFESSIONAL SUMMARY

Software Engineer with 7 years of experience building and operating web platforms end to end: backend in Node.js (NestJS) and Python (FastAPI), frontends in Next.js, React and Angular, and the AWS infrastructure they run on. Comfortable owning a feature from data model to deployment, with a track record of reducing latency and cloud costs in production systems. Work with AI every day: integrate LLMs into products (agents, RAG) and use Claude Code and LLM tooling to deliver faster. Apply Clean Architecture and TDD.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, Bash
Backend: Node.js, NestJS, Express, FastAPI, Spring Boot, REST APIs, Microservices, WebSockets, RabbitMQ, BullMQ, Clean Architecture, TDD
Frontend: Next.js, React, Angular, Mapbox GL, Leaflet, MapLibre
Databases & Search: PostgreSQL, pgvector, MySQL, MongoDB, Amazon DocumentDB, Redis, Elasticsearch, Firebase (RTDB, Firestore)
Cloud & DevOps: AWS (EKS, ECS, Lambda, RDS, DocumentDB, S3, CloudFront, CloudWatch), Kubernetes, Helm, Terraform, ArgoCD, Docker, GitHub Actions, CI/CD, Nginx, Linux, GCP, Azure
AI & LLM: LangChain, LangGraph, OpenAI API, Google Gemini API, Anthropic Claude API, AWS Bedrock, AI agents and tool calling, RAG, embeddings, Langfuse, MCP, n8n, Claude Code
Quality & Observability: Jest, Pytest, Elastic Stack, CloudWatch, Uptime Kuma, Git, Code Review, Scrum

WORK EXPERIENCE

Software Engineer (Full Stack & DevOps)

MapVX (Lazarillo) | Remote | Dec 2021 - Present

- Own the MapVX indoor-mapping platform end to end: the public APIs, the web apps and the editor the content team uses to publish each venue (NestJS, FastAPI, Next.js).
- Redesigned indoor routing so each building is routed in memory from a compact graph instead of a 44-million-document precomputed cache, taking route latency below a millisecond and removing the platform's largest database cost (DocumentDB).
- Took geospatial search and route precomputation from several seconds to under one by fixing query and index design, the slowest step for kiosks and the mobile app (PostgreSQL, Elasticsearch).
- Cut Firebase database egress by about 70% after tracing a billing spike to whole-index reads, and left a watchdog that alerts the team before a spike becomes an invoice.
- Launched the MapVX conversational assistant, which lets visitors find stores and routes by chatting on WhatsApp or the web, with fallback across LLM providers so the service stays up if one fails (LangChain, LangGraph).
- Made production deployments routine: moved legacy servers onto Kubernetes, codified the three AWS accounts in Terraform and replaced manual releases with CI/CD, with no downtime for clients.
- Keep the platform observable and secure: monitoring and incident response, 85+ sites behind a reverse proxy with automated TLS, and a WordPress compromise eradicated and rebuilt with zero downtime.

Backend Developer

ExmerDev | Loja, Ecuador | Sep 2020 - Nov 2021

- Delivered the roles, users and security modules of a trading and CRM platform and decoupled its services with asynchronous messaging so teams could release independently (Spring Boot, RabbitMQ).
- Sped up the platform's database queries and documented its REST API, cutting integration time for the frontend and mobile teams (MySQL, PostgreSQL, MongoDB).

Backend Lead

Teaching Innovation and Research Lab, Universidad Técnica Particular de Loja | Loja, Ecuador | Apr 2019 - Aug 2020

- Digitized the lab's attendance, people and materials management with a REST API that replaced a fully manual process (Node.js, PostgreSQL).
- Led the backend of the LiiD Manager platform (health and education) and set the team's technical standards and code review practice.

EDUCATION

Bachelor of Science (B.S.) in Computer Systems Engineering

Universidad Técnica Particular de Loja | Loja, Ecuador | 2017 - 2022

Relevant coursework: Artificial Intelligence, Databases, Operating Systems, Software Engineering, Algorithms and Data Structures, Computer Networks.

LANGUAGES

Spanish (Native) | English (Professional working proficiency)